http://en.wikipedia.org/wiki/Calculus_of_variations. For a few technical details about the Professor's equation, see the appendix *Some Technical Notes*.

Chapter 26. Singularities and Quantum Gravity

For a first foray into quantum fluctuations and quantum physics more generally, I recommend *The Ghost in the Atom: A Discussion of the Mysteries of Quantum Physics* (Davies and Brown 1986). I don't know any articles or books for nonphysicists about the quantum behavior of human-sized objects such as LIGO's mirrors; at a technical level, I discuss this in the second half of my third Pauli lecture (the one listed first) at http://www.multimedia.ethz.ch/speakers/pauli/2011. In John Wheeler's autobiography, he discusses how he came up with the idea of quantum foam (Chapter 11 of *Geons, Black Holes and Quantum Foam: A Life in Physics* [Wheeler and Ford 1998]).

In Chapter 11 of *Black Holes & Time Warps* (Thorne 1994) I discuss what was known in 1994 about the interiors of black holes, and how we came to know it—including the BKL singularity and its dynamics; quantum gravity's control of the singularity's core and its connection to quantum foam; and the infalling singularity (mass-inflation singularity), which had only recently been discovered by Erik Poisson and Werner Israel (Poisson and Israel 1990) and was not yet fully understood. The upflying singularity was discovered so recently that there is not yet any detailed discussion of it for nonphysicists; the technical discovery article is Marolf and Ori (2013) by Donald Marolf and Amos Ori. Matthew Choptuik's discovery that tiny, transient naked singularities are possible was announced and explained in his technical article (Choptuik 1993).